

Pacific Biomedical Engineering Network

6TH PACIFIC BIOMEDICAL ENGINEERING NETWORK (PBEN) MEETING REPORT

9th & 10th April 2024, Tanoa International Hotel, Nadi, Fiji

Prepared by Pacific Community

Pacific Biomedical Engineering Network

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Annex 2: List of Participants..... 20

Meeting papers are available at [here](#). PBEN recommendations are in [Annex 1]. The list of participants is in [Annex 2].

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Meeting Objectives:

1. The objectives of the meeting are for member countries to share Biomedical experiences, challenges and to strengthen partnerships towards addressing Biomedical priorities in Pacific Island Countries & Territories (PICTs). Topics of interest include:
 - **Capacity building:** Strengthen knowledge and skills through training. The workshop will include technical training of biomedical test equipment, servicing, maintenance, and trouble shooting.
 - **Network:** Support biomedical services and skills through strong collaboration and Partnerships
 - **Leadership:** Strengthen leadership and management skills for qualified biomedical technicians targeted towards country's needs.
 - **Gender Equality, Disability Social Inclusion (GEDSI):** Increase awareness on GEDSI matters in the health profession and biomedical field.

Opening and Devotion

2. Opening remarks were provided by Dr Silina Motofaga, Team Leader of the Clinical Services Team, on behalf of the Director of the Public Health Division, Dr Berlin Kafoa. The opening prayer was given by Mr Ambrose Kwaramb from Papua New Guinea. Dr Motofaga then welcomed participants and provided a brief overview underscoring the multi-dimensional approach to health innovation and collaboration in the Pacific region. She made mention the Pacific Heads of Health meeting in Samoa in April where a key feature will be a biomedical paper presentation, the first in a long while, showcasing significant findings and recommendations from a recent situation biomedical analysis study. She stated that the Pacific Biomedical Engineering Network meeting aims to outline key result areas (KRAs) linked to the broader vision and mission of fostering a unified Pacific community, addressing Development Goal 3 specifically and it will cover various technical and experience spheres within public health. Dr Motofaga highlighted five key result areas shared across different programs, including Clinical Services, NCD, SPRP (Surveillance, Preparedness, and Response Program), Lab Program, Health System Strengthening, and a newly added Climate Change Adviser role. She also encouraged participants to share ideas throughout the meeting, formally and informally, and make use of the opportunities to connect with colleagues across the region.

Session 1.2 Update on Regional Biomedical Services

Presented by: Ms Sunema Talapusi, Biomedical Adviser, Clinical Services Programme, SPC

3. The transition to "Pacific Biomedical Engineering Network" reflects a broader focus beyond biomedical services to include engineering aspects.
4. The importance of the Pacific Biomedical Situational Analysis 2023 report as a pivotal data source for decision-making and policy formulation was highlighted.
5. Key achievements at the 5th PBEN Meeting included the first certified technical training for the participants.

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6. Four in-country missions to Nauru, Cook Islands, Solomon Islands, and Samoa from August 2023 – March 2024, were undertaken to address specific biomedical needs ranging from policy formulation and implementation to technical training in biomedical workshop organisation, documentation management, and equipment maintenance. These missions were crucial in enhancing the workflow efficiency and overall biomedical service delivery in these countries.
7. The establishment of a technical subcommittee, tasked with drafting biomedical papers to aid in policy formulation and standard implementation across the region. The capacity building efforts by SPC involve extensive training programs focusing on electrical safety, ventilator training, and biosafety training to ensure up-to-date skillsets for biomedical technicians.
8. Procurement challenges were tackled by introducing international standards and tools like Emergency Care Research Institute (ECRI) for equipment and performance analysis, significantly easing the equipment procurement process. Additionally, the provision of laptops and desktops was made to facilitate better documentation and record keeping within the member countries.
9. The pressing need for commitment from both the country members and their directors to overcome challenges such as slow responses and lack of follow-through on commitments. Lack of data or outdated information, which affects support and informed cost-effective decisions was also highlighted.
10. **Recommendations:**
 - i. Continue to strengthen the PBEN network.
 - ii. Continue to support country requests for medical equipment procurement and/or technical training.
 - iii. More country missions to the North Pacific Islands & follow-up visits to other member countries and where required.

2.0 Workforce

Session 2.1 Mentoring & In-country mission

Presented by: Mr Tociro Katsuki & Mr Semisi Finau, DFAT Biomedical Consultants for Tonga & Tuvalu

11. The speakers shared their experiences and insights from their assignment to the Solomon Islands, particularly their support during the XVII Pacific Games. They were part of a team tasked with assisting in the installation of medical equipment at a clinic designed to provide special treatment to athletes. They expressed gratitude for working alongside the biomedical engineering team in Solomon Islands and mentioned the resolution of issues encountered with the assistance of SPC's biomedical adviser.
12. The presentation touched on broader challenges common to the participants, emphasising leadership, and the role of biomedical engineering services in improving healthcare. The speakers stressed the importance of continuous improvement, competent asset management, and preventative maintenance to avoid equipment failures. They highlighted issues related to insufficient working space, lack of proper testing equipment, and the need for thorough documentation and reporting as key areas requiring attention.
13. One significant technical challenge discussed was the fault detected during the installation of equipment due to improper electrical grounding, which posed a safety risk. The need for proper

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- testing equipment to ensure patient safety and equipment performance is emphasised, illustrating the problem with a specific example involving an air compressor and water removal.
14. The importance of having a visionary leadership approach, encouraging team inclusion, and leveraging individual strengths were discussed. The speakers also stressed the necessity of financial and logistical creativity to overcome resource limitations, sharing success stories in asset management and workspace organisation.
 15. The presentation concluded with a discussion on dew point issues in World Health Organisation (WHO) donated oxygen plants across several Pacific countries such as Tonga & Solomon Islands, advising on choosing between desiccant and refrigeration dryers based on local conditions.
 16. **Recommendations:**
 - i. Training in proper use of electrical safety testing and reporting.
 - ii. Training on documentation and reporting.
 - iii. Ensure all testing procedures and protocols are carried out such as acceptance and performance verification testing.
 - iv. Training on proper procurement and ordering of equipment and spare parts.
 - v. Refresher training for the end users and daily maintenance of equipment.
 - vi. Require assistance in developing and implementing of medical equipment policies.
 - vii. Guidance in the use of medical equipment standards and regulation
 - viii. Recommendation for improvement of Biomedical structure and establishment to accommodate new graduates and technicians.

Session 2.2 Challenges in biomedical HR provision at remote location

Presented by: Mr Oswald Malisa, Biomedical Technician, Vanuatu

17. A comprehensive overview of the biomedical technician infrastructure and challenges in Vanuatu, focusing on the Northern Provincial Hospital. Vanuatu consists of 82 islands divided into 6 provinces, each with a hospital, but only 2 main hospitals exist. The biomedical human resource situation is currently under strain with only 2 biomedical technicians in the country and one trainee; notably, the country's sole in-country biomedical engineer is nearing retirement.
18. The presentation details the responsibilities and reach of a biomedical technician at the Northern Provincial Hospital, emphasizing the management of over 1,000 medical equipment and the provision of maintenance services to three additional provincial hospitals. Furthermore, there's a stark depiction of the biomedical support challenges across the provinces, illustrating the scarcity of biomedical technicians, inadequate budget allocations, and logistical issues due to Vanuatu's geographical setup.
19. Gratitude was expressed for the support received from various sources, underlining the importance of collaborative efforts in overcoming the sector's challenges.
20. **Recommendations:**
 - i. Establishing a Biomedical Position within the Clinical services delivery relating to assets management in the Ministry to address the biomedical Issues in each Hospital.
 - ii. Provide basic biomedical hands-on training for suitable maintenance staff in each hospital as person of contact.
 - iii. Quarterly support/Training and assistance provided to provincial hospitals.
 - iv. All Biomedical Technicians and Engineers to be in Permanent Post.

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- v. MOH Vanuatu to put more emphasis on biomedical Department in terms of human resources.

Session 2.3 Questions and Answers

- 21. Papua New Guinea (PNG) emphasised on the importance of data security in asset management systems and explore sustainable, cost-effective solutions.
- 22. SPC commented on the utilisation of Microsoft Excel for asset management due to cost considerations but acknowledged its security limitations.
- 23. Fiji shared their experiences on challenges and solutions in managing medical equipment effectively.
- 24. Tonga stressed on electrical safety and the necessity of verifying power point voltage before using medical equipment to ensure patient and equipment safety.
- 25. PNG discussed the need for proper training, adequate equipment-to-medical personnel ratios, and geographical considerations in equipment procurement.
- 26. SPC encouraged adherence to a donation policy that aligns with the country's needs to prevent unnecessary burdens from equipment donations.
- 27. Tonga advocated for strong leadership that fosters good listening and effective group discussions for the collective improvement of biomedical practices.

3.0 Education

Session 3.1 Certified Technical Training

Presented by: Ms Ataata Binataake, Biomedical Engineer, Kiribati

- 28. The training focused on three key areas: electrical safety testing, biosafety cabinet training, and Hamilton ventilator technical training through an online course. The training has led to significant improvements, including the implementation of electrical safety testing for new medical equipment and the servicing and certification of biosafety cabinets. The team managed to secure crucial equipment for testing, such as an electrical safety analyser and an oxygen analyser, improving the safety and operational standards of medical equipment maintenance within their country.
- 29. One notable achievement is the understanding and application of safety testing to newly acquired medical equipment from Japan, where around 20 pieces were tested successfully. This is a first for Kiribati using the analyzer and adopting the train the trainer style from the certified training conducted in the 5th PBEN meeting in 2023.
- 30. Information sharing and practical learning have been paramount, helping improve the Kiribati biomedical team capabilities and service delivery standards.
- 31. The unique benefits of the Hamilton ventilator training, highlighting its preference among clinicians despite limited availability. The comprehensive training covered preventive maintenance, service software calibration, and provided a deeper understanding of ventilation technology.

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32. Challenges such as the scarcity of testing equipment, the need for more training to improve confidence among team members in safety testing, and recalibration issues were candidly discussed. The presentation also touched on the importance of external support in building capacity and ensuring the sustainability of these initiatives.
33. **Recommendations:**
- Exploring support for capacity building for the whole team to allow them to master the skills.
 - Exploring support from development partners for the provisional of unavailable test equipment.
 - Incorporating the electrical safety testing in preventative maintenance of all equipment.
 - Motivating the local team to stay committed in implementing the electrical safety testing and accurate completion of the Inspection and Acceptance forms.
 - Equipping the team with the proper tools for efficient data recording.

Session 3.2 The Medical Room Training

Presented by: Ms Sera Vereadali, Head of Biomed Department, CWMH Suva and Ms Virisila Livicala, National Biomedical Coordinator, FPBS Fiji

34. Clinical drug training, emphasising its alignment with Fiji's National Development Plan aimed at improving healthcare quality. The training, a collaboration between various partners including FEMAT, SPC Biomedical, and the Fiji Ministry of Health and Medical Services, underscores the growing importance of biomedical engineering in managing healthcare technology and patient health in Fiji.
35. Details of the training analysis process, identifying skills gaps among biomedical technicians through self and supervisor assessments. This analysis led to the realisation of a need for both short-term and long-term training to fill these gaps, with a particular emphasis on a short-term approach to immediately address the urgent needs.
36. The strategic planning and execution of training programs for biomedical technicians, catered by CureKids and supported by SPC and the Ministry of Health and Medical Services, showcasing the collaborative effort to upscale the workforce. The training, involving a mix of technical skills ranging from biomedical safety to the management of various medical equipment, resulted in the enhancement of participant skills, as evidenced by post-training analysis.
37. The outcomes of the training, including the establishment of monitoring groups, evaluation of biomedical testing equipment, and recommendations for ongoing training and retraining to keep pace with technological advancements. The commitment to fostering collaboration with the private sector and regional partners is also highlighted as a key step towards sustaining the growth and development of the biomedical engineering field in Fiji and beyond.
38. **Recommendations:**
- Establish a monitoring and compliance working group within the MOH for each member country.

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- ii. Post training monitoring on all biomed staff who attended technical training and to monitor the application of their new acquired skills and knowledge. It is important to also monitor whether there is a change in their practices to develop the biomed unit further 6 months to a year after training.
- iii. Purchase of Biomedical Test Equipment.
- iv. Re-training programme for a sustainable biomedical engineering
- v. Establish the Training Database – for skill set and competency.
- vi. Increased collaboration in Fiji with private sector and the region.

Session 3.3 Questions and Answers

- 39. American Samoa questioned the effectiveness of post-training motivation for local technicians and shared an example of a pay rise incentive from his workplace.
- 40. Fiji representatives discussed current strategies for technician retention, mentioning both monetary and non-monetary motivation, emphasising the significance of employee satisfaction, preventive maintenance skills, and mindset changes for managing biomedical equipment. Fiji's approach to securing support from the Ministry of Health and Medical Services comprises tying proposals to the Ministry's goals, conducting training needs analyses, and proving the need for continuous retraining due to the evolving nature of biomedical engineering.
- 41. Kiribati emphasised on patient safety as a motivational factor due to close-knit community ties and the challenges of short-term training incentives are highlighted.
- 42. Nauru also agrees with the pressure from clinicians preferring the Hamilton ventilator over the other brands having the same problem of having only one donated medical equipment. The biomedics have limited time to perform preventative maintenance because of the demands from the clinicians. It was then agreed by the PBEN that if a new equipment is donated, it is recommended to have a minimum of two equipment, to have a backup medical equipment.
- 43. SPC advised leveraging regional recommendations to advocate for training incentives at the national level.
- 44. PNG stressed the importance of having structured training pathways, policy inclusion, and the provision of allowances to support motivated staff, while also pointing out the challenges of translating training into productivity.
- 45. Samoa inquired about securing ministry support prompts advice on persistence and alignment with organisational objectives for securing funding or support.

Session 3.4 Fiji National University experience

Presented by: Mr Tuifonualava Tangatailoa, Head of Biomed, Tonga

- 46. Experiences and challenges encountered while undertaking a certificate course in Biomedical Engineering at Fijian National University (FNU) in 2003 were discussed.
- 47. The course, spanned across four quarters, each lasting eight weeks, and comprised six units focusing on electronic, electrical, and biomedical equipment, including their functions and electrical safety tests. The objective of the course was highlighted, which was to prepare and certify students for specialisation in Biomedical Engineering.

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48. Despite the course's value, challenges such as the intense workload, lack of proper equipment and safety standards, and absence of a conducive workshop for training were encountered.
49. **Recommendations:**
 - i. Recommendation for Diploma level at FNU to be negotiated by CMNHS.
 - ii. Proper testing equipment to be available for FNU course.
 - iii. Continuous support for training from donor agencies and SPC.
 - iv. PBEN to investigate certifications of biomedical technicians in the Pacific Islands.
 - v. PBEN to review the Certificate IV courses at FNU.
 - Any improvement or recommendation for the course.
 - Is the course appropriate and fit for what we do in biomedical engineering field?
 - vi. Licencing of Biomedical Equipment Technician (BMET).

4.0 Leadership

Session 4.1 Biomedical Services in Papua New Guinea (PNG)

Presented by: Mr Ambrose Kwaramb, Chief Engineer, Department of Health, PNG

50. The Biomedical Engineering Service's establishment within the government hierarchy, emphasised its vital role in supporting clinical services through infrastructure and engineering functions.
51. The presentation highlighted the key enablers including the National Health Plan 2021 – 2030 and National Service Standards, which guide the service's alignment with broader health system goals.
52. A historical overview of the biomedical engineering field's evolution, from the 1970s' focus on x-ray technology to the current scope, which encompasses intricate medical equipment and technologies.
53. Challenges faced were due to increasing population and technological advancement, along with the essential role of technical training and the recent initiation of a bachelor's program in biomedical engineering to address workforce development needs.
54. Partnership and collaboration between government, academia, industry, and donors are identified as fundamental frameworks for enhancing service capabilities and ensuring sustainability.
55. A vision for the future, focusing on continuous improvement and strategic strengthening in policy, provision, products, personnel, and partnerships to build a resilient biomedical engineering service.
56. **Recommendations:**
 - i. Create biomedical engineers/technicians' positions at NDOH, PMGH and PHAs.
 - ii. Develop a:
 - proper industrial training program for biomedical engineers/technicians.
 - graduate development programs for biomedical engineers.
 - competency-based training development plan.
 - national mentoring program for biomedical engineers and technicians.

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Session 4.2 Biomedical Services in Solomon Islands

Presented by: Mr Douglas Rerese, Principal Biomedical, Solomon Islands

57. The presentation highlights the progress and development of the biomedical unit under the Australian Aid Pacific Island Project (PIP) since its inception in 1998 and official recognition in 2006.
58. Key milestones included the formulation of a draft policy for medical equipment management in 2016 and reviewed again in 2023. The biomedical team, consisting of skilled engineers and technicians, supports 10 hospitals serving over 750,000 people across various islands and provinces.
59. Significant achievements included the drafting of policy, installation of a new PSA plant funded by World Health Organisation (WHO), development of an asset registry by a graduate biomedical engineer, and the proposed biomedical unit structure approved by the Ministry of Health.
60. The unit faces challenges such as inadequate funding, control measurements on equipment, and infrastructure limitations.
61. The concerted effort of the biomedical engineering department in collaboration with international stakeholders like the Japanese government, SPC, and the WHO to enhance healthcare services and capacity building among local technicians are acknowledged.
62. **Recommendations:**
 - i. Reviewing and updating the medical equipment management policy. This includes a need for the Ministry of Health to implement the generic SPC donation policy.
 - ii. Administrative assistance from SPC to gaining influence and support from our local government and ministries to understand our role and the importance of effective equipment management.
 - iii. To ensure that all provinces will have their own biomedical engineer who can provide advice and technical support to their health services and facilities.
 - iv. Providing initiatives and trainings to improve our young technician's technical and professional skills especially in electrical safety testing, documentation as well as getting involve in the procurement and policies development.

Session 4.3 Questions and Answers

63. Samoa asked the length of time for orientation for new hire to get up to speed with the current environment.
64. PNG emphasised on the two-year resident biomedical program for new hires to adapt to the professional environment, focusing on hands-on training in their facilities. Partnerships with industry players like Siemens and GE for specific product training. Highlighted the pathway for professional development post the initial program, leading to opportunities for obtaining a master's degree and higher advancements within the field.
65. Samoa acknowledged the lone female that graduate from the Biomedical program in PNG.
66. Tonga acknowledged the need for international collaboration for training programs with reputable institutions and organisations globally to enhance the skill set of the biomedical workforce.

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67. SPC discussed the integration of technical training within the broader health policy and strategic framework to ensure alignment and efficiency in workforce development efforts.
68. PNG addressed challenges in job placement and description post-certification, urging the need to align educational programs and training with market needs and job opportunities in the field of biomedical engineering.
69. Solomon Island in response to Fiji's question on Asset Registry stressed the importance of establishing a coherent and functional asset management system for medical equipment, aiding in the optimisation of healthcare services through proper equipment planning and management.
70. SPC suggested improving coordination between clinical services and biomedical equipment planning to ensure that the healthcare service delivery is efficiently supported by the biomedical engineering sector.

5.0 Preventative Maintenance & Asset Inventory

Session 5.1 Implementation of policies to informed decision making

Presented by: Ms Catherine Fiu, Principal Biomedical Engineer, Samoa, and Brandon Schuster, Biomedical Engineer, Samoa.

71. The presentation focused on preventative maintenance in the healthcare sector, specifically focusing on the implementation of policies and informed decision-making based on past preventative maintenance trips. It delved into the importance of preventative maintenance, which includes increasing the lifespan of medical equipment, identifying potential problems before they occur, enhancing communication with medical stakeholders, ensuring equipment safety, and providing refresher training for users.
72. The background provided on Samoa's healthcare infrastructure highlighted the challenges of managing numerous medical equipment across several district hospitals. The team's accomplishments in routine maintenance encompassed quality assurance, identifying, and addressing clinical support, education, and management, and analysers for preventive maintenance testing. Emphasising quality assurance, they discussed the importance of feedback and identifying service gaps during maintenance processes, incorporating team audits, and a yearly review to enhance service provision to hospitals.
73. Preventative maintenance challenges identified included inefficient asset management, lack of technical training for Biomedical Engineering (BME) staff, lack of user implementation, and disposal of hazardous materials.
74. Solutions proposed to overcome these challenges focused on implementing an improved asset management system, encouraging training collaborations across Pacific Island nations, reviving the Medical Equipment Management Committee, and developing a medical equipment waste policy.
75. **Recommendations**
 - i. Refresher trainings for nursing and clinical staff both off-site and on-site training on medical equipment. These refresher trainings will re-emphasize the importance of proper handling of medical equipment for the safety of the patient and the user. These training will also prolong the lifespan of the medical equipment which will be cost-effective.

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- ii. Encouraging Pacific Island Training Collaboration where biomedics are exchanged for a short period between member countries to be exposed to different types of biomedical practices whether it be documentation, corrective and preventative maintenance etc in each country.
- iii. The need and importance to revive the Medical Equipment Management Committee so that medical equipment policies and procedures can be implemented and supported by each country.
- iv. To incorporate medical equipment waste in current healthcare waste policies and procedures and to discuss with the relevant healthcare waste stakeholders.

Session 5.2 Documentation & Asset Inventory

Presented by: Mr Nelson Vaguna, Biomedical Technician III, American Samoa

- 76. American Samoa has a 150-bed hospital equipped with over 1,200 pieces of equipment. Three key aspects of asset inventory are highlighted: the provision of identity records for each piece of equipment, the management of equipment data and information capacity for efficient tracking, and the valuation of equipment to aid in forecasting and budgeting for purchases and maintenance.
- 77. The ongoing issue of maintaining proper condition reports for equipment, stressing the lack of proper documentation for more than a decade since 2013 and its implications are highlighted. Proper documentation is crucial as it provides evidence of workload, justifying daily operations, addressing equipment issues, and fulfilling auditing requirements.
- 78. The Hospital's transition to digital asset management through the implementation of a Computerized Management Maintenance System named 'micromanage' facilitates access to documentation, safeguarding against data loss, and protecting against unauthorised access.
- 79. An offer to share insights and potential benefits of the Asset Management System with other institutions or Pacific Island countries lacking this technology.
- 80. **Recommendations**
 - i. For PBEN to identify a suitable and reliable asset inventory which is economical and readily available for PBEN countries. The asset inventory should be able to update the PM schedules, location and other essential biomedical tasks to make informed decisions especially relating to procurement.
 - ii. Developing and upgrading skills on Micromains system focusing on Medical Equipment.
 - iii. Improve the sharing and transferring of information.

Session 5.3 Questions and Answers

- 81. Fiji asked on the timeline for implementation and challenges of a digital asset management for biomedical equipment in Samoa.
- 82. Samoa stated that implementation is set to start in July 2024, with a four-month period for equipment labelling and adjustments, focusing first on converting paper forms to digital and

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enhancing security protocols. The system will be national and not limited to biomedical equipment but aims to include all assets.

83. Several country participants shared their experiences and challenges with the Tupaia app, citing its complexity, security concerns and the need for customisation. Issues with testing equipment due to incompatibility with existing analysers and the necessity for certification are also discussed.
84. American Samoa shifted focus on preventative maintenance, highlighting discrepancies in equipment management, and the inefficiency of dual maintenance systems.
85. Suppliers Device Technologies suggested developing a standard operating procedure (SOP) with a decision-making process chart for managing equipment servicing, to streamline communication between clinical staff and the biomedical team.
86. SPC suggested Pacific Island countries to include formalising processes, emphasising the importance of contractual agreements for consumables and spare parts during procurement, and potential bulk ordering of consumables by Pacific Island countries to tackle consistent supply issues.
87. Suppliers Wishmed encourage collaboration between countries for bulk purchasing of consumables and spare parts to mitigate supply challenges.
88. SPC encouraged involvement in equipment procurement processes to include clauses for installation, user training, technical support, and consumable supply for the lifespan of equipment.
89. Suppliers Medica Pasifika suggested implementing forecasting for consumables based on patient load estimates to ensure a constant supply and prevent equipment inoperability.
90. SPC encouraged collaboration between Pacific Island countries for bulk purchasing of consumables and spare parts to mitigate supply challenges.

Recap of Day 2 – Key Points & Recommendations

Presented by: Ms Virisila Livicala, National Biomedical Coordinator, FPBS Fiji

91. Key PBEN recommendations from Day 1 of Meeting proper are in Annex 1.

6.0 Medical Waste

Session 6.1 Country experience & Biomedical Waste Management

Presented by: Mr Alben Adelbai, Biomedical Equipment Specialist, Palau

92. The Republic of Palau has one hospital with a 100-bed capacity.
93. The biomedical engineering team accomplishments included the opening of a new ICU building donated by JICA, a new MRI accepting patients, and the tele-endoscopy program, which is the second of its kind globally, allowing remote guidance of procedures via Zoom.
94. The core issue of medical equipment waste management was the challenge of recycling, noting that medical equipment contributes to pollution with limited recycling options. Only 12% of waste is recycled in Palau, with medical equipment waste constituting a small fraction.

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95. Insights into decommissioning processes for medical devices, emphasised the recyclability of certain components like wires and steel. The difficulties of achieving total recycling due to cost and lack of expertise, and the unprofitability and environmental hazards of exporting waste.
96. **Recommendations:**
- i. Proper care for Medical Equipment ensures long service life, therefore, less, or delayed contribution to waste.
 - ii. Careful processing of Medical Equipment to collect sought after recyclable components.
 - Copper, steel, brass, and lead.
 - iii. Purchasing of High-Quality Medical Equipment.
 - Greater vendor or manufacture support.
 - Longer product life support.
 - Lasts longer than cheaper lower quality equipment.

Session 6.2 Questions and Answers

97. Samoa questioned about the processing of medical parts and who undertook this responsibility, highlighting a practice observed in the Cook Islands involving outsourcing to a New Zealand company. Also, the operational protocols and practices involved in dealing with medical equipment – from waste management to decommissioning and purchasing strategies.
98. Palau revealed that, unlike the example provided, the presented organisation does not outsource this task but rather relies on a national waste processing centre and an internal maintenance department for handling medical equipment waste. They also engage in disassembling critical equipment like ventilators for part salvage and recycling, a practice beneficial for training new hires and promoting good practice. On purchasing decisions for medical equipment, the emphasis was placed on choosing high-quality products from reputable countries and using tools like ECRI for comparison, underlying a strategic approach favouring long-term quality.
99. Kiribati asked about decommissioning medical equipment and if the team gets authorisation before decommissioning.
100. Palau stated that approvals from management and justifications for both purchasing and decommissioning equipment were discussed, highlighting procedural necessities.
101. Fiji asked if Palau had a waste management policy in place. Concerning the disposal protocols for components such as generators and tubes from microscopy machines, Palau mentioned that no set protocol exists. However, they ensured that environmental hazards like oil and metals within these components are properly communicated to their national waste management facility.
102. SPC suggested a practice of sending out donation requirements to ensure donated equipment meets the organisation's needs.

Session 6.3 Group Discussions

- **What are the common causes of medical equipment waste in PICT?**
 - Poor or lack of implementation of the donation policy resulting in unwanted/unplanned/unexpected and obsolete medical equipment donations
 - Inactive or no waste management policy

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- Lack of consultation of biomed during procurement leading to poor quality of medical equipment procured, expensive spare parts for maintenance, expensive consumables and accessories. Also, equipment not suitable for the Pacific context.
- Damaged equipment by poor handling by operators
- Equipment has reached end-of-life and therefore replacement is cost effective than repairing as it is beyond economical repair.
-
- **Was it always an issue or is this a new issue?**
 - Majority of the group stated that it has been a long term unresolved issues due to non-compliant donations, damaged medical equipment and has accelerated due to the influx of new equipment during Covid-19.
- **What is the healthcare waste management process in your countries? Is it effective?**
 - Not as effective as it should be but recommended for collaboration with SPREP, Ministry of Environment, Ministry of Works and other stakeholders is important in managing the healthcare waste effectively and for its safe disposal.
 - Public Health and Environmental Act
 - Medical waste usually go to MOH incinerators
 - General waste to landfill
 - Waste containment and segregation at the point of waste generation (green bin for general waste, yellow bin for infections waste, blue bin for other hazardous waste)
 - Cook Islands – existing Waste Management department
- **Is there a healthcare waste policy? Who are the stakeholders?**
 - Yes, there are some countries with Healthcare waste policy such as Fiji & PNG - Waste Management Division, Public Health, City Council. They work together with the Ministry of Environment and SPREP in implementing policies. However, there is no clause specifically for medical equipment waste. Therefore, medical equipment waste is not recognised / addressed in the existing environmental acts or laws in Fiji such as National Solid Waste Management Strategy 2011-2014 & National Waste Strategy 2011-2015
 - PNG Enforcement Act, IPC Policy, Healthcare Waste Management Policy under development (draft) with support from SPREP.
 - However, the PBEN from Kiribati, Niue, Nauru, Samoa, and American Samoa are not aware of a healthcare waste policy from their respective countries.
- **What recommendations in addressing healthcare waste?**
 - Enforcement of existing policies
 - Waste Management Policy
 - Environment Acts
 - MOH must actively monitor the proper disposal of medical equipment
 - For Member Countries to conduct the following:
 - Review existing policies with all relevant stakeholders.
 - Identify health stakeholders for collaboration.
 - Develop healthcare waste management policy and develop specific medical equipment waste management policy including E-waste
 - Conduct training for hazardous waste awareness and BME to take part in it.
 - Explore Waste Company organization
 - Separating usable parts and use it as spare parts for other in-service equipment
 - For Fiji to review existing Drug Waste Management Policy (FPBS)

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7.0 IPC Hand Hygiene

Session 7.1 Hand Hygiene workshop – Practical

Presented by: Ms Margaret Leong, Infection Prevention & Control Adviser, SPC

103. A detailed educational session focused on Infection Prevention and Control (IPC) for healthcare workers, regardless of their role or location within a hospital. The aim was to discuss, interactively, the significance of IPC, the chain of infection, and how infections spread, focusing particularly on standard and transmission-based precautions, hand hygiene, cleaning and disinfection of patient care equipment.
104. A video detailing real-world examples to illustrate healthcare-associated infections and their transmission in healthcare settings, emphasising the high transmission rates and consequences, including death.
105. Global issues of IPC were highlighted with WHO statistics showing that 1 in 10 hospital patients develop a healthcare-associated infection, which can lead to increased morbidity and mortality rates due to antibiotic resistance. The discourse also touches on the alarming fact that up to 90% of healthcare workers don't practice proper hand hygiene, underscoring the crucial role of effective IPC programs in reducing these infections by up to 70% according to the latest WHO report.
106. Examples from the Pacific, particularly Fiji, are shared, including outbreaks in neonatal units due to IPC breakdowns, pointing out the drastic outcomes of inadequate hand hygiene, equipment reuse, and insufficient cleaning/disinfection practices.
107. The importance of breaking the chain of infection for COVID-19 through isolation, vaccination, social distancing, and wearing PPE were discussed. Highlighting the necessity of applying standard precautions in all patient interactions regardless of their health status to prevent infection spread.
108. The presentation stressed the significance of hand hygiene in preventing healthcare-associated infections and outline the correct durations for hand washing and sanitization based on whether hands are visibly dirty or not. Provided evidence for the prescribed hand hygiene durations and how they effectively reduce the bacterial load on hands, showcasing the comparative effectiveness of soap and water versus alcohol-based hand rubs.

Session 7.2 Group Discussions

109. An interactive activity asking participants to apply their knowledge by mapping out how COVID-19 fits into the chain of infection, encouraging them to think about each link and how to break it as a measure to prevent infections.
 - **Choose 3 common equipment that are shared on patients and determine how they should be cleaned and stored after use.**
 1. Patient monitor – wipe down with a cloth.
 2. Ventilator – specific cleaning solution. Alcohol solution.

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3. Infusion pumps – wipes
- **List 5 high tough points that could be classified as potentially contamination in a patient's room.**
 1. Bed rails
 2. Door handles
 3. Equipment with screens and keyboards
 4. Power Points
 5. Appliances

110. An activity involving hand hygiene practices among healthcare workers, reinforcing the critical role of IPC measures in healthcare settings.

- The prescribed durations for hand washing (40-60 seconds with soap and water) and hand sanitization (20-30 seconds with hand rub) are based on their effectiveness in reducing bacterial load, as presented through evidence.

8.0 Discussion of Recommendations

Session 8.1 Key Decision Points & Recommendations Country Representative Only (CLOSED DOORS)

Session 8.2 Key Decision Points & Recommendations

Capacity Building:

- i. Suggestion for future biomedical training to include American standards for relevant member countries such as American Samoa, Palau and other North Pacific Island countries.
- ii. Biomedical training and retraining program with retention plan for sustainable biomedical services as there has been a lot of migration of skilled technical workforce to developing countries.
- iii. Recommendations for development partners to include narrative on support for training, workforce capacity building, and funding.
- iv. Emphasise on co-funded initiatives to ensure country ownership and alignment to workforce planning.
- v. Member countries to support the sharing of biomedical expertise between countries in the region for exposure on biomedical equipment management practices on operational and management level.
- vi. Enforcement of the donation policy to avoid increased biomedical waste and non-compliant medical equipment. Awareness to clinical stakeholders on the donation policy with its checklist for effective implementation as donations should not incur operational cost for the recipient country.
- vii. Biomed to become a mandatory member of clinical service networks within the regions to enable a cohesive and sustainable service delivery.
- viii. Review the existing healthcare waste policy to include medical equipment waste management and safe disposal.
- ix. Standard biomedical engineering workshop specification for scalability to each country's context.
- x. Adequate and dedicated biomedical budget for medical equipment management support.

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- xi. Procedure to approach SPC for support aligns requests to workforce plans and may include co-funding.

Closing

The closing prayer was provided by American Samoa.

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Annex 1: Key Points & Recommendations from Day 2 of PBEN – Day 1 of Meeting Proper



Recap of PBEN Day 2
- Key Points & Recom

Annex 2: List of Participants



Participant List - 6th
PBEN Final.pdf